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The integer 120 has many factorizations. For example, 120 = (2)(60), 120 = (3) (4)(10), and 120 = (–1)(–3)(4)(10). In how many of the factorizations of 120 are the factors consecutive integers in ascending order?

A. 2

B. 3

C. 4

D. 5

E. 6

•

Jorge's bank statement showed a balance that was \$0.54 greater than what his records showed. He discovered that he had written a check for $\$x.yz$ and had recorded it as $\$x.zy$, where each of x, y, and z represents a digit from 0 though 9. Which of the following could be the value of z?

A. 2

B. 3

C. 4

D. 5

E. 6

•

One side of a parking stall is defined by a straight stripe that consists of n painted sections of equal length with an unpainted section $\frac{1}{2}$ as long between each pair of consecutive painted sections. The total length of the stripe from the beginning of the first painted section to the end of the last painted section is 203 inches. If n is an integer and the length, in inches, of each unpainted section is an integer greater than 2, what is the value of n?

A. 5

B. 9

C. 10

D. 14

E. 29

•

This year, Henry will save a certain amount of his income, and he will spend the rest. Next year, Henry will have no income, but for each dollar that he saves this year, he will have $1 + r$ dollars available to spend. In terms of r, what fraction of his income should Henry save this year so that next year the amount he has available to spend will be equal to half the amount that he spends this year?

A. $\frac{1}{r} + 2$

B. $\frac{1}{2r} + 2$

C. $\frac{1}{3r} + 2$

D. $\frac{1}{r} + 3$

E. $\frac{1}{2r} + 3$

	Consecutive Minutes	Units of Power

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Machine	Consecutive Minutes Machine Is Off	Units of Power When On
A	17	15
B	14	18
C	11	12

•

At a certain factory, each of Machines A, B, and C is periodically on for exactly 1 minute and periodically off for a fixed number of consecutive minutes. The table above shows that Machine A is on and uses 15 units of power every 18th minute, Machine B is on and uses 18 units of power every 15th minute, and Machine C is on and uses 12 units of power every 12th minute. The factory has a backup generator that operates only when the total power usage of the 3 machines exceeds 30 units of power. What is the time interval, in minutes, between consecutive times the backup generator begins to operate?

A. 36
B. 63
C. 90
D. 180
E. 270

•

In a certain region, the number of children who have been vaccinated against rubella is twice the number who have been vaccinated against mumps. The number who have been vaccinated against both is twice the number who have been vaccinated only against mumps. If 5,000 have been vaccinated against both, how many have been vaccinated only against rubella?

A. 2,500
B. 7,500
C. 10,000
D. 15,000
E. 17,500

•

Three boxes of supplies have an average (arithmetic mean) weight of 7 kilograms and a median weight of 9 kilograms. What is the maximum possible weight, in kilograms, of the lightest box?

A. 1
B. 2
C. 3
D. 4
E. 5

Stock	Number of Shares
V	68
W	112
X	56

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Y	94
Z	45

•

The table shows the number of shares of each of the 5 stocks owned by Mr. Sami. If Mr. Sami were to sell 20 shares of Stock X and buy 24 shares of Stock Y, what would be the increase in the range of the numbers of shares of the 5 stocks owned by Mr. Sami?

A. 4

B. 6

C. 9

D. 15

E. 20

•

Last year, sales at Company X were 10% greater in February than in January, 15% less in March than in February, 20% greater in April than in March, 10% less in May than in April, and 5% greater in June than in May. In which month were sales closest to the sales in January?

A. February

B. March

C. April

D. May

E. June

•

If s and t are integers greater than 1 and each is a factor of the integer n, which of the following must be a factor of n^{st}?

I. s^t

II. $(st)^2$

III. $s+t$

A. None

B. I only

C. II only

D. III only

E. I and II

•

How many 4-digit positive integers are there in which all 4 digits are even?

A. 625

B. 600

C. 500

D. 400

E. 256

•

If $0 < r < 1 < s < 2$, which of the following must be less than 1?

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If $0 < r < 1 < s < 2$, which of the following must be less than 1?

I. $\frac{r}{s}$

II. rs

III. $s - r$

A. I only

B. II only

C. III only

D. I and II

E. I and III

•

Last month, 15 homes were sold in Town X. The average (arithmetic mean) sale price of the homes was $150,000 and the median sale price was $130,000. Which of the following statements must be true?

I. At least one of the homes was sold for more than $165,000.

II. At least one of the homes was sold for more than $130,000 and less than $150,000.

III. At least one of the homes was sold for less than $130,000.

A. I only

B. II only

C. III only

D. I and II

E. I and III

•

Pumps A, B, and C operate at their respective constant rates. Pumps A and B, operating simultaneously, can fill a certain tank in $\frac{6}{5}$ hours; Pumps A and C, operating simultaneously, can fill the tank in $\frac{3}{2}$ hours; and Pumps B and C, operating simultaneously, can fill the tank in 2 hours. How many hours does it take Pumps A, B, and C, operating simultaneously, to fill the tank?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

E. 1

•

If n is a positive integer, then $(-2^n)^{-2} + (2^{-n})^2 =$

A. 0

B. 2^{-2n}

C. 2^{2n}

D. 2^{-2n+1}

E. 2^{2n+1}

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•

Which of the following is equal to $\left(\frac{2\sqrt[3]{3}}{\sqrt{2}}\right)^3$?

A. $\frac{3\sqrt{2}}{2}$

B. $3\sqrt{2}$

C. $6\sqrt{2}$

D. 12

E. $12\sqrt{2}$

•

Dara ran on a treadmill that had a readout indicating the time remaining in her exercise session. When the readout indicated 24 min 18 sec, she had completed 10% of her exercise session. The readout indicated which of the following when she had completed 40% of her exercise session?

A. 10 min 48 sec

B. 14 min 52 sec

C. 14 min 58 sec

D. 16 min 6 sec

E. 16 min 12 sec

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